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Review and reframing of the operationalisation of the water planetary boundary for river basin planning

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**Review and Reframing of the Operationalisation of the Water Planetary
Boundary for River Basin Planning**

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Abstract

The Planetary Boundaries framework defines a Safe Operating Space for humanity by
setting biophysical boundaries to critical Earth System components, including the water
cycle. This framework is a promising paradigm to address the challenges posed by
increasing anthropogenic pressures and climate change impacts on water systems.
Nevertheless, the Water Planetary Boundary framework has not yet been systematically
integrated into basin-scale water management planning. We address this challenge by first
reviewing the development of the global Water Planetary Boundary concept and
summarising the approaches used to apply this concept at sub-global scales, while
highlighting its merits and gaps. Building on this review, we argue that existing research on
operationalising the Safe Operating Space for water at sub-global scales often examines
limited variables, neglecting a comprehensive basin-scale approach and failing to provide
adaptable frameworks for other regions or future scenarios. At the same time, the global

Water Planetary Boundary framework, although setting aspirational goals, lacks actionable implementation strategies. We propose a novel framework for operationalising the Water Planetary Boundary principles at the basin scale, integrating socio-ecological functions into the assessment of the basin-scale water boundary. Our framework is based on the assumption that operationalising the Water Planetary Boundary principles at the basin level requires prioritising Earth's stability within a Holocene-like state and social equity. This involves preserving water functions critical to ecosystem services and ensuring the resilience of water systems for current and future generations. The framework's application can enhance the understanding of how human activities interfere with the water cycle and highlight critical issues that must be addressed to maintain the water system within a Holocene-like state at the basin scale.

Introduction

The stability of the Earth System in the Holocene, i.e., the last 11,000 years, is arguably at the origin of the agricultural revolution, which initiated the great human civilisations of the past and ultimately evolved into contemporary society (Feynman and Ruzmaikin 2007). Therefore, the thriving of human society is inextricably linked to the biophysical conditions that enabled its rise and development. Central to this stability is a predictable water cycle, which acts as a cornerstone of the Earth System, essential for sustaining ecological integrity and societal functions (McMillan et al., 2025). However, the growing development of economies and societies worldwide has amplified human pressures on water resources, with global water withdrawals increasing sixfold over the last century (Kahil et al. 2015; Wada et al. 2013, 2016). Moreover, anthropogenic climate change exacerbates this strain by increasing temperatures, altering precipitation patterns, intensifying hydrological variability, and undermining the predictability of water availability (IPCC 2001). These disturbances to the water cycle, combined with the ever-increasing demand for water driven by growing food and energy demands and improved living standards, further intensify water management challenges (Firoozi et al., 2026). These include creating conflicts within and between countries (Jiang et al., 2025; Oki and Quiocho, 2020; Veldkamp et al., 2017), escalating water scarcity and heightening the frequency and intensity of extreme events such as floods and droughts (McCoy et al., 2022; Rohde, 2023), as well as causing groundwater depletion, degradation of aquatic ecosystems, and loss of biodiversity (Bierkens and Wada 2019; Bunn and Arthington 2002; Gallardo et al., 2018; Pastor et al. 2014; Tickner et al. 2020).

Innovative paradigms and technical solutions for managing water resources in a changing climate and society are urgently needed to ensure sufficient and reliable water quantity and quality for functional natural ecosystems supporting human activities in the coming decades. One scientific paradigm that can guide these transformations and has gained considerable traction recently is the "Planetary Boundaries" (PB) framework introduced by Rockström et al. (2009a; 2009b). The PB concept outlines how humanity can achieve sustainable development by operating within a Safe Operating Space (SOS) defined by biophysical boundaries, thereby preventing disruptive anthropogenic interference with critical Earth System processes (Scheffer et al. 2015; Steffen et al. 2015). Applied to

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freshwater, staying within the SOS means ensuring that freshwater resources can be used to meet human needs (e.g., food, drinking water or energy) without negatively affecting the present and future functioning of water systems and their resilience (Rockström and Karlberg 2010). Although some attempts have been made to adopt the PB framework for regional and local decision-making (Hjalsted et al. 2021) it has not yet been systematically integrated into basin-scale water management planning. Moreover, no structured and transferable methodology is available to effectively and holistically operationalise the PB framework in practical water management contexts. We see a significant opportunity to improve basin-scale decision-making by developing regional applications of the global PB framework (Stewart-Koster et al., 2024). This can be achieved through water management decisions that account for the functional relationships and trade-offs among critical components of the water system. Operationalisation must also address the evolving nature of the global PB concept, which has progressed as advances in scientific understanding and technical capabilities have enabled more precise and sophisticated assessments. As such, there is an opportunity to develop a flexible and robust framework for operationalising the PB principles in water resource management. This paper aims to seize this opportunity by (i) reviewing the development and current state of knowledge of the PB water concept at the global scale, (ii) reviewing and summarising existing approaches used to apply the concept, drawing on the global related concepts, highlighting merits and gaps, and (iii) proposing a new basin-scale operationalisation of the PB to guide water management plans.

The Planetary Boundaries Framework

The concept of a Safe Operating Space (SOS) for humanity was originally introduced within the Planetary Boundaries (PBs) framework by Rockström et al. (2009a; 2009b). This framework applies critical transitions theory (Scheffer et al. 2001) to establish boundaries for key biophysical state variables of the Earth System, using the Holocene as a reference period. The PBs framework identifies nine critical Earth System processes: climate change, freshwater use, nitrogen and phosphorus cycles, ocean acidification, chemical pollution, atmospheric aerosol pollution, stratospheric ozone depletion, biodiversity loss, and land-use change (Rockström et al. 2009b). A control variable and a response variable define each of the nine PBs. The control variable is a measurable quantity representing the state of an Earth System process. The response variable reflects how the Earth System reacts to changes in the control variable (Bunsen et al. 2021; Gleeson et al. 2020b). Control variables have intrinsic thresholds, i.e. critical tipping points, that, if crossed, can push the Earth System away from the Holocene-like state, potentially causing abrupt environmental changes, ecosystem degradation, and adverse impacts on human societies (Steffen et al. 2015). In contrast, PB are precautionary, human-defined limits set at a “safe” distance from those thresholds. They are shaped by societal judgments about acceptable risk and the uncertainties surrounding the exact location of thresholds. While thresholds are inherent to the Earth System and reflect its biophysical properties, boundaries are normative and externally imposed to guide human activity. The two concepts are thus distinct yet interlinked: understanding thresholds is essential to establishing effective boundaries, while adherence to boundaries can help prevent the crossing of critical thresholds and

maintain Earth System stability. This Holocene-like stability remains the only known state of the Earth capable of supporting complex human societies.

The PB framework also distinguishes between boundaries with global thresholds and those with regional-scale thresholds. Boundaries with global thresholds—such as atmospheric CO₂ concentration—have immediate and widespread consequences when exceeded, though impacts may vary regionally (Steffen et al. 2015). In contrast, regional-scale thresholds—such as water overexploitation — can interact with other Earth System processes and, when crossed globally, cumulative local impacts can trigger cascading effects that extend to global repercussions, ultimately threatening planetary stability and human well-being (Rockström et al. 2009a).

The PB framework was later integrated with social considerations, emphasising the need to ensure universal access to essential social foundations for human well-being. The social foundation is defined by eleven dimensions: water, income, education, resilience, voice, jobs, energy, social and gender equity, health and food (Raworth 2012). Although none of these dimensions are explicitly linked to individual PBs, maintaining the social foundation inherently involves resource use, implying a minimum level of natural resource consumption below which human deprivation occurs. Raworth (2012) visualised the integration of PBs and the social foundation using a doughnut-shaped model, where the outer ring represents the environmental ceiling (i.e. the PBs) and the inner ring marks the social foundation. The space between the two rings represents humanity's "safe and just" operating space. The doughnut concept is founded on the assumption that social injustice and inequalities create conditions unsuitable for maintaining a functional environment. Conversely, adverse environmental conditions caused by trespassing of the PB environmental ceiling can drive societies below the social foundation. In the doughnut concept, PB and social justice are interdependent: achieving sustainability is essential for justice, but environmental policies must safeguard human well-being (Raworth 2012).

This integrated vision has recently evolved into the Safe and Just Earth System Boundaries (ESBs) framework (Rockström et al. 2023), incorporating justice criteria into the PB framework. The ESBs aim to ensure that planetary stability coexists with equitable access to essential resources, such as water, food, energy, and infrastructure (Gupta et al. 2023; Rockström et al. 2023). The Earth System Boundaries framework introduces three justice criteria: (1) interspecies justice, to protect biodiversity by avoiding transgression of PBs; (2) intergenerational justice, recognising the need to preserve Earth System stability for future generations; and (3) intragenerational justice, addressing inequalities among countries, communities, and individuals to minimise harm caused by one group to another. By complying with these criteria, the Earth System's stability is preserved, and thus species are conserved, and future generations are protected (Rockström et al. 2023).

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Methodology

Search Strategy

A systematic search was conducted across the “Scopus” and “Web of Science” databases to identify peer-reviewed articles linking the concepts of Safe Operating Space or Planetary Boundaries with basin-scale water management. The search was conducted using four search strings to capture a wide range of relevant terminologies and framings. To ensure comprehensive coverage of the foundational theoretical concepts, well-known seminal papers from the authors of the original Safe Operating Space framework were manually added to the review selection, in addition to those identified through the systematic search.

A total of 169 peer-reviewed articles were identified using four search strings; this initial list was pruned for duplicates using the Cadima web platform (<https://www.cadima.info/>), yielding 84 unique records. A second round of exclusion was performed based on the titles, abstracts, and contents of the remaining papers, yielding 36 papers. The criteria for inclusion in this list were: a) the paper must deal at least partially with the water planetary boundary global or local definition; b) the paper must deal with the operationalisation or downscaling of the water planetary boundary to the local scale. In addition to the papers thus selected, a total of four papers were added to the corpus based on personal knowledge because of their high importance but published after the search was conducted.

The search strings and the pruning steps are documented in the supplementary material as search_strings.pdf and search_results_filtering.xlsx, respectively.

Results

Evolution of the Water Planetary Boundary

The Water Planetary Boundary (WPB) has evolved from a global, consumption-based metric to a more nuanced framework considering spatial variability, environmental flow requirements, and the intricate relationships between water and other Earth System components (Figure 1). Initially defined as a global limit on blue water consumption, the WPB has been refined to incorporate green water and acknowledge water scarcity's local and regional impacts. The control variables used to define the WPB have also shifted over time. Early iterations focused on global water consumption and the withdrawal of blue water. However, later studies recognised the importance of including spatially explicit Environmental Flow Requirements (EFR) and the distinction between global and basin-scale boundaries. More recent approaches emphasise the need to consider the interactions between the water cycle and other Earth System components, using a range of indicators such as soil moisture, streamflow, and groundwater levels to assess the impact of water use on these systems.

Control Variables

The inclusion of water in the PB framework was justified by Rockström et al. (2009a; 2009b) based on the argument that cumulative local and regional water threshold breaches can contribute to catastrophic global consequences. The WPB was initially defined as 4,000

1 km³/year of global consumptive blue water use, reflecting water withdrawals that are not
2 returned to the system. This measure was chosen not to enforce a cap on global water use
3 but due to limited knowledge of more suitable indicators (Rockström et al., 2009b), thus
4 acting as a proxy for human perturbation of the water cycle. Recognising water systems'
5 spatial and temporal variability, Gerten et al. (2013) revised the WPB by integrating EFR
6 using ensemble hydrological models. This approach considered the fraction of streamflow
7 left untouched to maintain ecosystems while retaining blue water consumption as the
8 control variable. Critiques have noted that blue water overuse is typically confined to the
9 basin scale and may not always result in global-scale effects (Heistermann 2017). Steffen et
10 al. (2015) further refined the WPB framework, introducing regional boundaries alongside
11 the global ones. Regional WPBs for freshwater were defined using monthly streamflow
12 withdrawal limits in large basins. Still, critiques persisted regarding the unclear relationship
13 between control and response variables and the lack of a comprehensive understanding of
14 interactions at regional and global scales (Gleeson et al. 2020b). The most significant
15 conceptual shift occurred with Gleeson et al. (2020a), who advocated for abandoning single
16 indicators, such as consumptive blue water, in favour of a multi-indicator system that
17 reflects four core water cycle functions: hydroclimatic regulation, hydroecological
18 regulation, water storage, and transport. These functions represent five key water stores—
19 atmospheric, soil moisture, surface, groundwater, and frozen water—each associated with a
20 planetary sub-boundary. This multi-indicator approach was later simplified into two core
21 sub-boundaries: blue water (encompassing surface and groundwater) and green water
22 (encompassing atmospheric and soil moisture) to enhance outreach (Gleeson et al. 2020b).
23 These sub-boundaries were operationalised in Richardson et al. (2023) using streamflow
24 and root-soil moisture as control variables, benchmarked against Holocene variability.

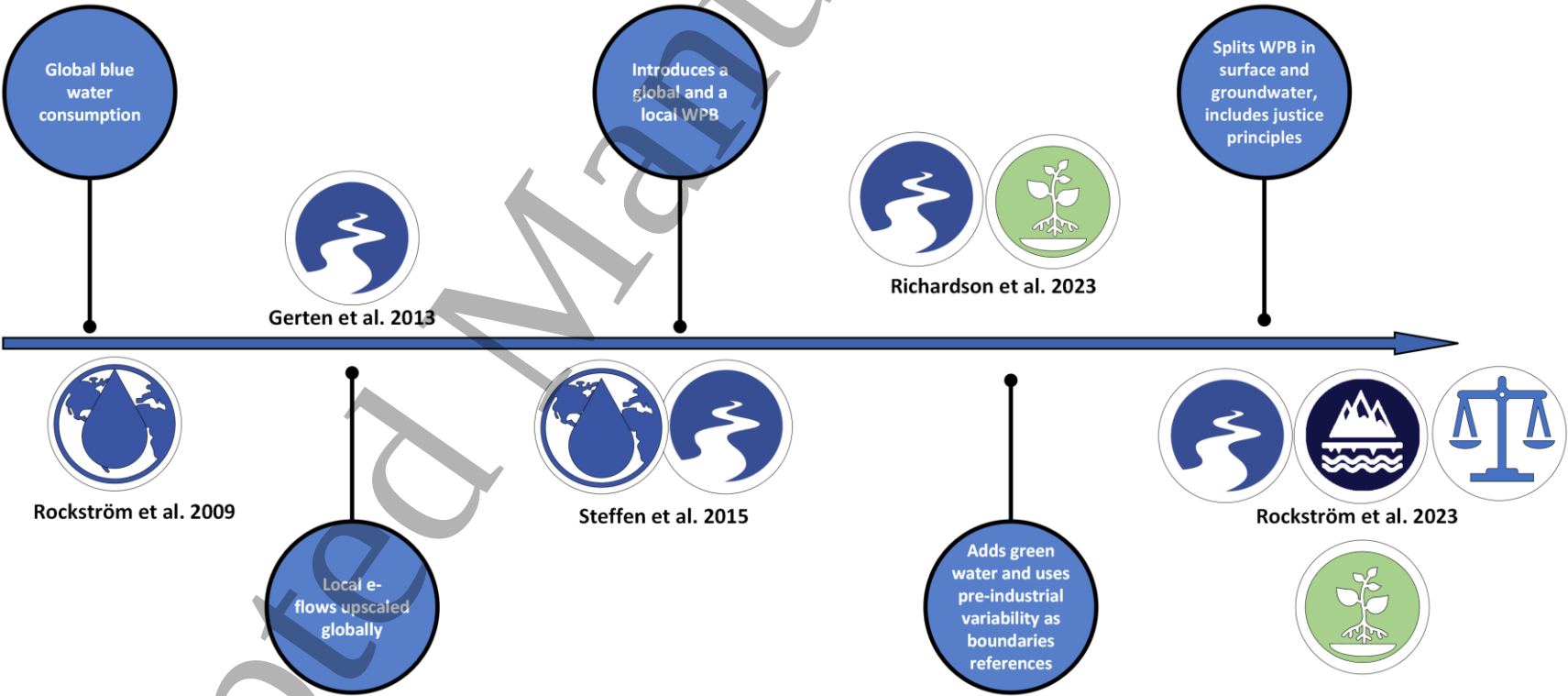


Figure 1: Water Planetary Boundary definitions timeline.

The revision also suggested a scale of analysis for each store consistent with the scale at which alterations to the store can yield globally relevant impacts. For example, basins with an area of between 10,000 and 100,000 km² for surface water and regional aquifers for groundwater (Gleeson et al. 2020a). Using sub-global WPB implies calculating these spatially distributed boundaries first and then aggregating them to calculate the cumulative global boundaries. Cumulative global boundaries can be computed with several approaches: i) an unweighted approach, i.e., summing up the percentage of global land over a given threshold; ii) a weighted approach, i.e., percentage of global land over a given threshold weighted by basin importance; iii) a keystone approach, i.e., giving higher importance for basins and aquifers of global relevance and; iv) using a tipping element approach, i.e., considering the water system factors contributing to maintaining Earth System elements of sub-continental relevance (e.g., monsoons) that, if lost would change planetary level shifts (Gleeson et al. 2020a).

Evolution of Thresholds

Threshold estimation has shifted alongside control variables. In Rockström et al. (2009a; 2009b), thresholds were implied through blue water consumption limits. Later iterations, including Gerten et al. (2013) and Steffen et al. (2015), refined thresholds using hydrological models, flow seasonality, and EFRs but struggled to define how local breaches translate into global instability. Gleeson et al. (2020b) proposed defining thresholds based on deviations from water cycle functions that maintain the stability of the Earth System. Richardson et al. (2023) quantified thresholds using hydrological reconstructions of the pre-industrial period, defining the WPB as trespassed when more than 10–11% of ice-free land falls outside the pre-industrial ranges of streamflow and soil moisture variability.

The Safe and Just ESB framework (Rockström et al., 2023) further evolved thresholds, introducing three spatially explicit sub-boundaries for surface water (flow alteration), groundwater (drawdown), and green water (soil moisture). These were benchmarked at global and local levels: 80% of river flows must remain unaltered, groundwater withdrawals must not exceed recharge, and soil moisture must remain within Holocene variability for 90% of the ice-free land. Critically, the groundwater threshold was questioned because using current abstraction conditions as a baseline overlooks past overuse, resulting in biased thresholds that may exclude vulnerable populations while allowing wealthier regions to continue overusing resources (Cuthbert et al. 2024).

Evolution of Trade-offs

The trade-offs inherent to water boundary setting have increasingly come under scrutiny. Initial WPB definitions treated blue water consumption as a global metric, overlooking socio-ecological trade-offs at the basin scale. Later formulations (Gerten et al., 2013; Steffen et al., 2015) acknowledged seasonal and regional dynamics but did not explicitly account for trade-offs between human and ecological water uses. Gleeson et al. (2020a) emphasised the importance of understanding trade-offs between competing water cycle functions. For instance, increased water storage through reservoirs may enhance hydroclimatic regulation

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1 but reduce hydroecological integrity. The revised WPB framework enables the assessment
2 of such trade-offs by assigning boundaries to distinct water functions and stores.

3 The Safe and Just ESB framework (Rockström et al., 2023; Gupta et al., 2023) incorporates
4 social trade-offs into PB by embedding justice into boundary-setting. It includes: 1)
5 Interspecies justice, ensuring that ecological water needs are met; 2) Intergenerational
6 justice, requiring long-term sustainability; and 3) Intragenerational justice, addressing
7 current inequalities in water access. This justice dimension adds a socio-political lens to
8 trade-offs, requiring policies to balance ecological integrity with equitable access to water.
9 For instance, achieving surface and groundwater targets should not marginalise low-income
10 populations or regions historically affected by over-extraction (Cuthbert et al. 2024).

11 **Downscaling Approaches**

12 Human society's growing pressure on water resources and the negative impacts of climate
13 change necessitate adopting water management policies and techniques centred around
14 sustainable and just paradigms. It is arguably for these reasons that the PB framework
15 gained popularity, and several studies have downscaled the global concept to the local scale.
16 In this context, the local scale refers to any scale other than the global, i.e., states, regions,
17 cities, administrative units in general, and even corporations (Cole et al. 2014; Huang et al.
18 2020; Kulionis and Pfister 2022; Li et al. 2020). Downscaling is instead meant to define one
19 or more control-response variable pairs to delimit a sub-global SOS for water. The
20 methodologies for downscaling the WPB can adopt either a top-down (e.g. Fang et al., 2015;
21 Nykvist et al., 2013; Li et al., 2020) or bottom-up approach (Cole et al., 2014; Teah et al.,
22 2016). In the top-down approach, part of the WPB is allocated to the local scale on a fair-
23 share basis, i.e., by assigning a linear proportion of the WPB to a specific property of the
24 analysed local unit. For example, allocation can be calculated as a population-proportional
25 fraction of the WPB, thus calculating the water allocation per capita (Nykvist et al., 2013;
26 Fang et al., 2015; Hoornweg et al., 2016).

27 The scale – global to local – and the downscaling approach reflect on the scope and potential
28 of the WPB studies. Studies employing a top-down approach typically align with national or
29 global scales and focus on Earth System conservation, water management or environmental
30 sustainability. Conversely, bottom-up approaches are predominantly linked to single water
31 bodies or local scales, with a focus on resource-specific environmental management (Table
32 1). Also, the spatial scale of the studies in Table 1 reveals a gradient in their focus. Smaller-
33 scale studies, concentrating on individual water bodies or small areas, tend to emphasise
34 specific environmental management goals, such as water quality or fish stocks, often with
35 an emphasis on human Well-being. Conversely, as the spatial scale increases to the national
36 or regional level, the focus shifts to overarching water management and broader
37 environmental sustainability frameworks, often focusing on blue water consumption and
38 virtual water flows. At the largest, global scale, the primary focus consistently relates to
39 environmental/Earth System conservation, addressing issues like planetary boundaries,
40 climate stability, and the integrity of terrestrial and aquatic biospheres. When downscaling
41 occurs with a fair share approach, the water consumed by the local unit is typically

1 calculated on a consumption basis, meaning that all water consumed, regardless of its origin
2 (domestic or foreign), is included in the computation of water consumption at the local
3 scale (Bunsen et al. 2021; Han et al. 2023; Moberg et al. 2020). This approach highlights the
4 full environmental impact of consumption, encompassing the water used throughout the
5 entire supply chain, from production to transportation and consumption (Li et al. 2020;
6 Ryberg et al. 2021). By taking a broader, global perspective, the consumption-based
7 approach can reveal the water footprints associated with consumption patterns in a
8 particular region, shedding light on the water demands of imported products and the
9 potential impact on water resources in other regions (Hoekstra and Hung 2005; Mekonnen
10 and Hoekstra 2011). Ultimately, this reveals the indirect impact of water consumption and
11 highlights countries that contribute to water scarcity elsewhere due to their consumption
12 patterns (Zipper et al. 2020; Bunsen, Berger, and Finkbeiner 2021).

13 The bottom-up approach applies the PB framework to define locally relevant control and
14 response variables, thereby establishing a local SOS (McLaughlin et al. 2018; Zipper et al.,
15 2020). It then computes the amount of the control variable that is locally consumed. Locally
16 relevant control variables reflect the state of one or more locally relevant water bodies,
17 such as a river or an aquifer. In most studies (e.g., Fanning and O'Neill 2016; Teah et al.
18 2016; Vargas et al., 2017; Bjorn et al. 2020; Huang et al. 2020), the control variables chosen
19 for the local SOS mirror those used in global WPB assessments, although examples exist of
20 local SOS considering water quality (Dearing et al. 2014). The amount of locally consumed
21 variables is computed on a production basis, i.e. considering only water withdrawals and
22 consumption within the analysed local scale and including, for example, water used for
23 agriculture, industry, and domestic needs (Zipper et al., 2020), thus disregarding foreign
24 water from imports and services. The bottom-up approach enables the assessment of the
25 environmental impact on local water resources associated with the water used to produce
26 goods and services within the analysed region. Thus, it tracks the direct environmental
27 impact of freshwater use across various sectors, aiding in identifying local excessive water
28 use. Nevertheless, quantifying local control variables and boundaries can be challenging due
29 to the limited data availability. Therefore, involving communities through a bottom-up
30 approach can help assess these values and thresholds while considering local
31 socioeconomic and environmental conditions (Teah et al., 2016; Vargas et al., 2017). In
32 Zhang et al (2022), the two approaches are combined into a single framework where the
33 bottom-up aggregation provides the internal distribution within the Chinese provinces
34 based on ecological efficiency defined as product between natural resources consumption
35 and economic output. The top-down adjustment instead, provides the external constraint in
36 terms of population fair-share of the PBs, thus ensuring the total stays within the global
37 limit.

Table 1: Spatial scale, downscaling approaches, control and response variables of the reviewed studies. ↑ Bottom-up approach. ↓ Top-down approach. Focus
acronyms: WM: Water Management, EM: Ecological Management, ES: Environmental Sustainability, HW: Human Well-being, EESC: Environmental and Earth
System Conservation

Author	Spatial Scale & Approach	Control Variable	Response Variable	Focus
Bai et al., 2020	Single Water Body - ↑	Aquatic macrophyte biomass	Lake terrestrialization	WE, EM
Bjorn et al., 2020	Global/Regional/National - ↓	Blue water consumption	Products' SOS for water	ES
Bunsen, Berger, & Finkbeiner 2021	National - ↓	Blue water consumption	National SOS for water	WM
Carpenter et al., 2017	General/Methodological - ↑	Multidimensional	Local SOS for a water body	EM
Cole et al., 2014	National - ↑	Blue water consumption	Ecological functionality	ES & HW
Cooper et al., 2019	Local - ↑	Freshwater & sediment delivery, water body salinity	Fish catch and stock	ES & HW
Dearing et al., 2014	Local - ↑	Human pressures (e.g. hydroelectric production, untreated sewage)	Water quantity and quality	ES & HW
Fang et al., 2015	National - ↓	National water (blue & green) footprint	Environmental Sustainability Ratio	ES
Fanning & O'Neill 2016	National/Sub-national - ↑	Blue water consumption	Ecological functionality	ES
Gerten et al., 2013	Global - ↑	Blue water consumption	Earth System resilience and biosphere integrity	EESC
Gleeson et al., 2020ab	Global - ↑	Human water cycle modification	Climate stability, terrestrial and aquatic biosphere integrity, and carbon uptake	EESC
Han et al., 2023	National/Sub-national - ↓	Blue water consumption	Coupling Coordination Degree	ES & HW
Hoekstra & Hung 2005	National - ↓	International virtual water flows	National virtual water balances	WM
Hoornweg et al., 2016	Local - ↑	Blue water consumption, potable water availability	Potable water supply & index of embodied water consumption	ES
Huang et al., 2020	National - ↑	Blue water consumption	Water Scarcity Index	WM
Kulionis & Pfister 2022	Global to Local - ↓ & ↑	Blue water consumption	Ecological Condition of Aquatic Systems	ES
Li et al., 2020	Sub-national - ↓	Blue water consumption		WM
McLaughlin et al., 2018	Regional/Sub-national - ↑	Dry season stream flow	Ecological and biological impacts	EESC
Moberg et al., 2020	National - ↓	Blue water consumption	Ecological functionality of water ecosystems	ES
Nykvist et al., 2013	National - ↓ & ↑	Blue water consumption	Altered Water Cycle	ES
Ofir et al., 2023	Single Water Body - ↑	Water body level	Fish catch	EM
Richardson et al., 2023	Global - ↑	Streamflow deviation from preindustrial period	Freshwater's Earth system functions	EESC
Rockström et al., 2009ab	Global - ↓	Multiple (CO2, ocean acidification, etc.)	Ecosystems collapse, major shifts in moisture	EESC
Rockström et al., 2023	Global/Regional - ↑	Streamflow alteration, quality standards	Collapse of freshwater ecosystems	EESC & HW
Ryberg et al., 2021	Global to Local - ↓	Blue water consumption	Environmental sustainability	ES
Steffen et al., 2015	Global/Regional - ↓ & ↑	Blue water consumption/Monthly flow alteration	Aquatic biodiversity loss; regime shifts	EESC
Szabó et al., 2020	Single Water Body - ↑	Biological indicators	Water body ecological state	EM & WM
Teah et al., 2016	Local - ↑	Blue water consumption	Water availability	ES
Vargas et al., 2017	Single Water Body - ↑	Streamflow alteration	Ecosystem condition and service supply	EM
Zhang et al 2022	Sub-national - ↑	Blue water consumption	Environmental Unsustainability Index	ES
Zipper et al., 2020	Global to Local - ↓ & ↑	Watersheds or total river length within limits	Aquatic biosphere integrity	EESC & WM

The fair shares and consumer-based approaches are linked to the PB because they are relevant to water's global functions. However, globally defined control variables might not be the most suitable indicators for assessing local water conditions or guiding local water management. On the other hand, combining the local SOS and the producer approach is relevant locally, as it highlights the impacts and pressures on local water resources and is a useful tool for targeting management interventions. At the same time, applying the local SOS does not inform the local contribution to cumulative global-scale impacts. To harness the advantages of both approaches, Zipper et al. (2020) proposed a method to bridge and harmonise the two scales. The method is based on the water stores from (Gleeson et al. 2020b) and encompasses three cases. In the first one, a given water store is relevant only at one of the two scales. In this case, a single boundary exists, and harmonisation is unnecessary. In the second case, the control variables differ for the local and global SOS, possibly because the Earth System core functions sustained by the store vary between the local and global contexts. The resulting SOS for this store encompasses both local and global boundaries, reflecting the duality of the store's significance at both global and local scales. The last case occurs when local and global control variables are the same, but their safe boundary differs. The boundary choice shall prefer the more environmentally conservative option, ensuring the store is maintained within the SOS at both levels.

Discussion

Operationalising the WPB at the Basin Scale

Despite the long history of the WPB concept and the numerous studies that have attempted to downscale it, a systematic methodology to inform the design of basin-scale water management plans based on the PB framework has yet to be documented. One possible reason is that the lack of global water governance institutions hinders the operationalisation of the WPB at the basin scale, as there are no planetary-scale bodies to enforce global water management, making a single global boundary of limited practical use (Zipper et al., 2020; Gleeson et al., 2020b). Furthermore, the approaches described in the previous section are only weakly informative for the different decision-making scales, from global to local, especially regarding the latter. Nevertheless, to be effective, the WPB must be applied locally or regionally, where water management occurs (Pastor et al., 2022; Gleeson et al., 2020b). Tracking Earth System interactions requires localised monitoring, as water impacts depend on specific local and regional factors, and ecological consequences are scale-dependent. For example, small environmental transgressions might be invisible at larger scales but can cascade into threats to planetary stability (Gleeson et al., 2020b; Li et al., 2021). Moreover, incorporating WPB into existing watershed frameworks, such as IWRM and WFD, has the advantage of harmonising global science with local approaches (Zipper et al., 2020; Li et al., 2021).

We argue that at the local scale, the framework must support the holistic design of measures to restore, preserve, or potentially expand the SOS, thereby fulfilling an operational function. Nevertheless, most of the studies that attempted a local operationalisation of the SOS for water mostly focused on a single (Cooper et al., 2019) or

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fewer variables of the water system (e.g. Bai et al., 2020; Carpenter et al., 2017; Ofir et al., 2023; Szabó et al., 2020), and no holistic research, including all dimensions of the water system at the basin scale, has been put forward.

We suggest that operationalising the WPB at the basin scale must be centred around the PB's founding concept of maintaining the Earth System within a Holocene state while ensuring a just society. Thus, the operationalisation of WPB should translate into the overall management objective of safeguarding the water functions that underpin the delivery of water ecosystem services and the resilience of the water system for present and future generations. Nevertheless, in our view, the current paradigm underpinning the definition of local water boundary (Gleeson et al. 2020b, 2020a) is not sufficient to guide the formulation of basin management plans to implement the PB principles because it uses single control variables for each store, does not explicitly consider water quality, and does not include a systematic consideration of the social foundation or local stakeholders engagement as suggested by several author (e.g. Hayha et al. 2016; Vargas et al. 2018). Moreover, although nutrients technically pertain to the nitrogen and phosphorus PB, it is impossible to discount the appraisal of water quality in river basin management. A further limitation of the current paradigm is the formulation of local boundaries based on water stores and single control variables for each store, which may not adequately capture basin management features and dynamics. The multidimensional nature of the water resources system makes it difficult to represent local boundaries in two dimensions, as in the global PB framework. The position of the basin-scale boundary is, in fact, more likely to be defined by a combination of intertwined social, ecological, and hydrological parameters (Zipper et al., 2020).

To account for the social, hydrological, and ecological parameters, we propose defining basin-scale sub-boundaries based on the blue water functions outlined in Falkenmark et al. (2019) and Falkenmark and Wang-Erlandsson (2021). Water functions are subdivided into hydro-ecological, hydro-climatic, and hydro-social categories (Table 2) and can be naturally occurring, such as blue water regulatory functions, or created and maintained by humans, such as the supply for human consumption. These functions (Table 2) are the processes supporting the resilience of the water system, with water resilience defined as the role of water in maintaining a given, desirable state of water-dependent environments, water supply for human society, and moisture feedback to the rest of the Earth Systems (Falkenmark et al., 2019). This composite social-ecological system, encompassing the environment, society, and Earth Systems, recognises the interdependence between the biophysical and anthropic dimensions of the water system and ultimately implicitly adheres to an SOS definition that includes Earth Systems and a just society.

The functions of Falkenmark et al. (2019) strongly overlap with those of (Gleeson et al. 2020b). Nevertheless, while the latter proposes a set of purely biophysical functions supporting Earth System stability, Falkenmark et al. (2019) also extend into the social dimension, defining the water functions that support social-ecological resilience.

We envision the basin-scale SOS for water as a space delimited by a hierarchy of indicators, variables, and water functions, similar to the example in Figure 2. The framework includes a

hierarchy of functional sub-boundaries that represent the basin system's water dimensions, as well as an Integrated Water Modelling System (IWMS) that provides biophysical and social representations of the basin. Interactions among stakeholders connect the two components.

In the functional sub-boundaries hierarchy, each water function has a boundary associated with control and response variables. The response variable describes the extent to which the function is fulfilled. Each control variable is a composite value created by aggregating multiple indicators chosen to represent the basin-scale processes that contribute to fulfilling the function. The SOS for water at the basin scale is achieved when all functions are fulfilled to a minimum level marked by their boundary.

Table 2: Blue water functions upholding water system resilience. Adapted and extended from Falkenmark and Wang-Erlandsson (2019).

Store	Function	Type	Processes	Examples
Bluewater	Regulatory	Hydro-ecological	Discharge intra- and inter-annual variations, radiation reflection, and carbon cycling	Floodplain rejuvenation, channel planform maintenance, nutrient pulses, wetland carbon sink, glaciers' albedo
	Productive	Hydro-ecological and hydro-social	Aquatic and groundwater-fed terrestrial vegetation growth, crop growth	Phreatophytes forests and macrophytes habitats maintenance. Crops irrigation
	Storage	Hydro-ecological and hydro-social	Hydrological connectivity and sea level	Aquatic habitats maintenance
	Chemicals carrier	Hydro-ecological	Suspension and solution transport	Nutrients and pollution transport
	Supply	Hydro-social	Water withdrawal	Supply of water for human consumption, industry and energy production.

The top level of the hierarchy, represented by the water functions, enables the framework to maintain strong linkages with Earth System functioning and be widely applicable since these functions are present in every basin. Conversely, the lower-level indicators in the hierarchy should be tailored to the basin's properties. Ideally, the lower level of the hierarchy should be co-created with the basin's stakeholders to incorporate their water values and knowledge of the basin's key features. In this context, water values refer to the social, cultural, environmental, and economic benefits that water provides (Haileslassie et al. 2022). Understanding and recognising water values is crucial for decision-making

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1 because considering the full spectrum of water values can lead to more informed and
2 equitable decisions regarding water allocation and infrastructure development.

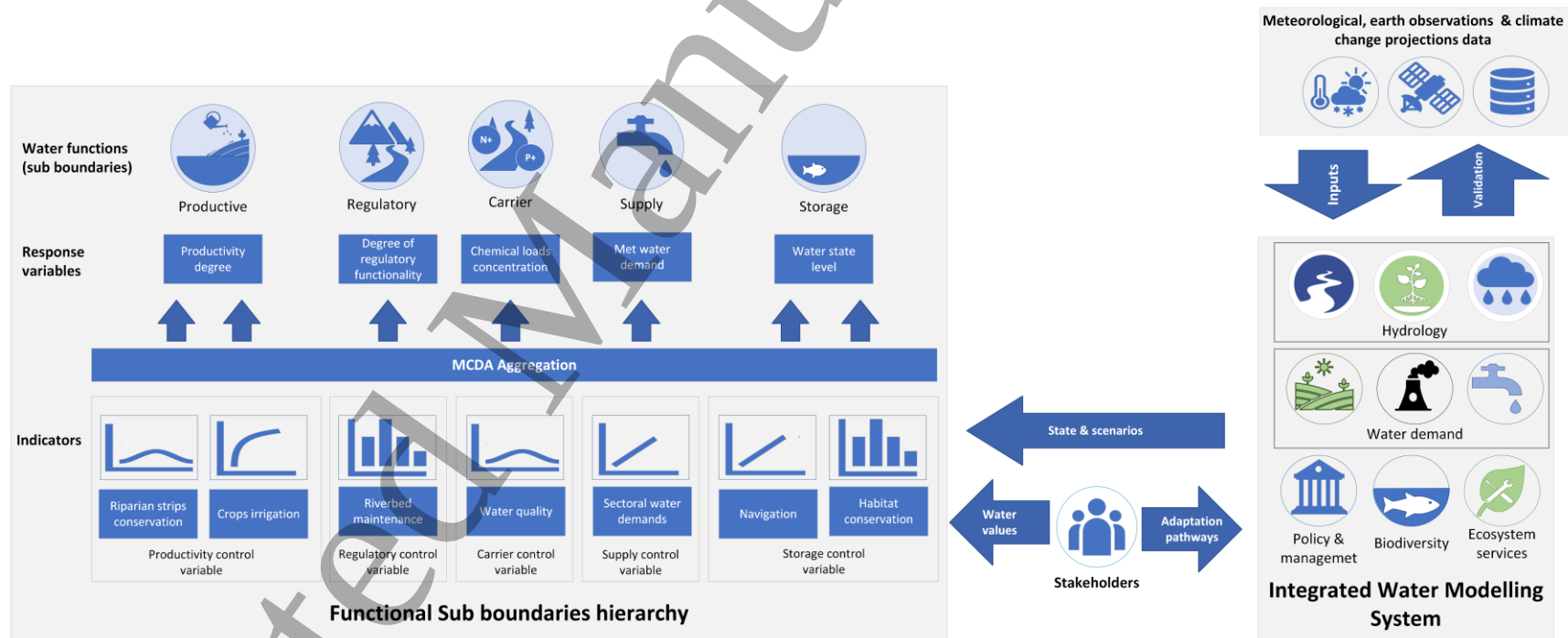


Figure 2 Framework for operationalising the SOS for Water at the basin scale. The functional sub-boundary hierarchy shows how a single or multiple indicators can comprise the control variables. The indicators are used to measure the basins' performance relative to the water values expressed by stakeholders. They are computed using the Integrated Water Modelling System, and their value must respond to simulated management decisions.

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For the framework to be applicable for assessing the SOS for water under current and future conditions, the data required to compute the indicators must be replicable by models composing an IWMS providing the socioecological and biophysical representation of the basin and its responses to policy-driven water allocations and socio-economic and climatic drivers. The IWMS must be capable of replicating the water cycle, human water demands, and biodiversity impacts, as well as implementing policy-driven rules and socio-economic drivers, thus providing a comprehensive tool to assess the current basin state and explore the solution space derived from alternative management pathways and climate and socio-economic change scenarios. Hydrological models (Arnold et al. 2012; Burek et al. 2020; Sutanudjaja et al. 2018), hydro-economic models (Kahil et al. 2015; Macian-Sorribes et al. 2015), biodiversity models (see Pollock et al. 2020 for an extensive review) and ecosystem services models (Sharp et al., 2018; Villa et al. 2009) are examples of the elements that can be used to build a holistic IWMS able to encompass all water dimensions relevant to basin management. Field data and remote sensing products are additional elements of the IWMS that can be used to set up and validate models against observations.

Once the indicators are computed, they must be aggregated to determine whether the basin is within an SOS. However, the choice of how to aggregate the indicators and setting the boundary value for each water function are not trivial tasks. Although water functions and indicators calculation are backed by science, aggregation and choice of boundary values are political processes involving decision-makers and stakeholders that must consider the trade-offs between scientific evidence and the burden required to maintain the basin within an SOS (Biermann and Kim 2020). Therefore, operationalising the SOS framework implies systematic conflict resolution to achieve consensus. While methodologies such as Participatory Rural Appraisal (PRA - Ahmed et al. 2019; Al-Qubatee et al. 2017), Delphi techniques (Lund et al. 2014), and focus group discussions (Restrepo-Osorio et al. 2022) excel at integrating local knowledge and structured expert insights, we argue that Multi-Criteria Decision Analysis (MCDA) via Multi-Attribute Value Theory (MAVT) is the most robust approach for the proposed SOS framework. MAVT's mathematical structure enables rational, multi-objective decision-making that withstands political and public scrutiny. Its empirical reliability is demonstrated by recent quantified evidence; for instance, a study of 100 stakeholders showed that MCDA produces consistent results across repeated opinion polls, successfully accounting for shifts in perspective over time (Kuller et al., 2023). This transparency and the ability to generate results that are explicitly justifiable to the public (Langhans and Schallenberg 2023; Reichert et al. 2013, 2015; Schuwirth et al. 2012) make MAVT particularly suited for assessing complex stakeholder preferences in water management. MAVT supports decision-making by combining quantitative data on the performance of alternatives across multiple criteria with stakeholder preferences that define value functions, weights, and trade-offs among them. This process is facilitated by developing an objectives hierarchy, which clarifies which aspects of the system should be assessed and why, ensuring that the most relevant dimensions of the decision problem are systematically considered. Consequently, MAVT applications typically rely on targeted stakeholder or expert samples rather than large-scale population surveys. Such samples generally include approximately 10 to 30 participants, purposefully selected to gather

informed preferences and value judgments relevant to the specific decision context (Scholten et al. 2017; Zheng and Lienert 2018). In some cases, however, this number may be higher when multiple stakeholder workshops are organised to examine how preferences evolve through iterative objective elicitation and ranking (Kuller et al. 2023). MAVT relies on value functions – a mathematical method for translating indicators' values into scores – which evaluate a vector representing various attribute levels, assess an option's impact, and convert it into a measure of how well objectives (indicators) are met, with a score ranging from 0 to 1, with higher scores indicating better performance (Haag et al. 2022). Each value function associated with an indicator is weighted on a consensus basis by stakeholders and decision-makers, typically gathered through interviews, workshops, or focus groups, and then used to construct the preference functions. The scores of the indicators related to the same water function are then combined using the functions' weights to determine the level of fulfilment of that water function.

A key challenge in applying multi-criteria decision analysis (MCDA) to river basin management lies in evaluating alternative pathways without inadvertently constraining the exploration of trade-offs among evolving multisectoral indicators and stakeholder preferences. Previous studies (e.g. Haimes and Hall 1977; Giuliani et al. 2014; Kasprzyk, Reed, and Hadka 2016) have shown that preferences shift when new objectives are introduced, often due to cognitive biases such as cognitive myopia, where narrowly defined optimality limits the discovery of decision-relevant alternatives that could reshape stakeholder preferences (Hogarth 1981)—and cognitive hysteresis, where conventional problem-solving strategies hinder the generation of novel, innovative decisions or additional objectives (Gettys and Fisher 1979). Addressing these biases, especially through the strong participation of stakeholders and decision-makers, is essential for ensuring that MCDA frameworks fully capture the complexity of local water resource management problems. Finally, when applying MCDA-MAVT or other stakeholder engagement approaches, it's crucial to ensure the actors representing basin interests adequately reflect diverse groups, including disadvantaged communities and environmental advocates. Water's shared, public nature and competing uses necessitate a suitable sample size to ensure inclusivity and support adherence to the Safe and Just ESB framework (Rockström et al., 2023; Gupta et al., 2023) principles of intergenerational, intragenerational, and interspecies justice.

The framework thus designed can be used to assess a basin's status and identify the most critical functions to preserve or restore. At the same time, using an IWMS allows for projecting the effects of climate change and policies into the future, informed by stakeholders' values. The integrated nature of the modelling system should also enable the identification of trade-offs resulting from different management choices, ultimately allowing assessment of whether the analysed basin will remain within a SOS for water.

Conclusion

We reviewed the evolution of the WPB definition from a global, consumption-based metric to a spatially variable and socio-environmentally aware limit and the methods and

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approaches used to apply the WPB to sub-global scales. Building on this knowledge, we propose a framework (Figure 2) that can be applied to any river basin, regardless of its size. The framework establishes a hierarchical approach anchored in core blue-water functions that are universally present across all basins, thereby ensuring a strong link to Earth System processes (Falkenmark 2017) and, by maintaining functionality within basin limits, preventing regional water overexploitation from triggering global cascading effects that threaten the WPB. The framework also emphasises operationalising Earth System justice by integrating stakeholder values and social-ecological functions to ensure that management decisions uphold social equity, biodiversity, and intergenerational and intragenerational justice, thereby supporting a stable, Holocene-like Earth system while promoting social fairness. The framework can assess whether the basin is within a Safe Operating Space for Water and project how the basin's status will change in response to climate change and management choices.

The framework's application relies on co-creating a functional hierarchy that describes the basin's functional components and the water values of its stakeholders. Additionally, the co-creation process fosters the public's involvement and buy-in in decision-making. Building the hierarchy around socio-ecological functions ensures that environmental objectives are considered, thereby preventing the maximisation of short-term economic benefits at the expense of long-term sustainability and implicitly incorporating elements of interspecies and intergenerational justice. At the same time, social-ecological functions also account for human needs, thereby avoiding management decisions that aim to maximise environmental objectives while disregarding societal well-being.

The suggested framework can support basin management by providing a structured system for holistically assessing the water system's functionality, considering trade-offs and synergies among water dimensions. This operationalisation can be further enhanced by integrating it with established frameworks for local-scale integrated water resources management. Since the pioneering Harvard Water Program (Lloyd et al. 1963), the field of Water Resources Systems Analysis has advanced significantly (see Brown et al. 2015 and references therein), leading to various decision-support frameworks focused primarily on adaptation design. These include decision scaling (e.g., C. Brown et al. 2012; Culley et al. 2016), robust decision making (e.g., Herman et al. 2015; Matteo Giuliani and Castelletti 2016; Zatarain Salazar, Castelletti, and Giuliani 2022), and dynamic adaptive policy pathways (e.g., Arnold et al. 2023; Haasnoot et al. 2013; Herman et al. 2020), all of which offer structured approaches for managing water resources at local scales amid uncertain future scenarios.

The framework we propose belongs to a type of management concept that aims to limit human interference in natural processes while recognising that environmental goals cannot be pursued without considering societal well-being (Zeyang et al., 2025; Calado et al., 2007). These approaches involve negotiating complex trade-offs between human economic development and ecological preservation and depend heavily on participatory, stakeholder-

1 driven political processes to define local environmental values, compile diverse scientific
2 indicators, and build consensus around sustainable management routes.

3 The proposed framework shares core similarities with well-established frameworks such as
4 Integrated Water Resources Management (IWRM) and the Water-Energy-Food-Ecosystems
5 (WEFE) nexus. All prioritise participatory, stakeholder-driven methodologies over top-
6 down management, emphasising decentralised decision-making and bottom-up co-creation
7 of solutions, and aim to balance human development with environmental sustainability and
8 social equity. They view water systems as integrated social-ecological systems and
9 prioritise equitable access to resources (Vrachioli et al., 2019; Meran et al., 2021). However,
10 the proposed framework and IWRM differ in their core focus: IWRM treats water as a scarce
11 economic resource, centred on efficiency and social welfare (Meran et al., 2021), while the
12 SOS framework is rooted in Earth System science, aiming to maintain biophysical stability
13 through Planetary Boundaries and multi-dimensional justice. This difference is reflected in
14 their application goals. Although both rely on complex, interdisciplinary modelling to
15 evaluate water system trade-offs, their core modelling logic differs: IWRM focuses on
16 economic optimisation to maximise aggregated social and economic benefits (Meran et al.,
17 2021); similarly, WEFE focuses on systemic efficiency, resource security, and economic
18 optimisation within the circular economy (Vrachi et al., 2019; Carmona-Moreno et al., 2025)
19 while the modelling we propose centres on maintaining functional sub-boundaries and
20 ecological thresholds to preserve biophysical stability, which in turn, is necessary to uphold
21 the ecosystems and their services that allow humans to thrive (also in economic terms) and
22 resource security. On the other hand, the Water-Energy-Food-Ecosystems (WEFE) nexus
23 and the Safe Operating Space (SOS) framework share the foundational principle of
24 balancing human development with environmental sustainability by treating water systems
25 as interconnected social-ecological entities (Vrachioli et al., 2019). Both paradigms reject
26 top-down decision-making in favour of participatory, stakeholder-driven methodologies—
27 such as co-design workshops and scenario-building—to capture local socio-economic
28 realities and "water values".

29 The major limitation of applying the framework is the high volume of data and modelling
30 capacity required to create an IWMS to assess whether a basin is in an SOS and to highlight
31 critical issues for current or future management decisions. While meteorological forcing
32 data (e.g., precipitation, temperature, etc.) are widely accessible from global datasets (e.g.,
33 W5E5-Karger et al., 2023), other hydrological variables such as discharge or groundwater
34 levels can be obtained through in-situ networks (e.g., GRDC: <https://grdc.bafg.de/> or United
35 States Geological Survey - USGS: <https://dashboard.waterdata.usgs.gov/>) and remote
36 sensing products like the Gravity Recovery and Climate Experiment (GRACE:
37 <https://gravis.gfz.de/>). However, the spatio-temporal resolution of hydrological variables
38 remains inconsistent worldwide; while North America and Western Europe benefit from
39 dense observational coverage, the Global South is comparatively data-poor (Burek and
40 Smilovic, 2023). Additionally, detailed data on anthropogenic water demands and
41 allocations are often unavailable, leading global models to rely on coarse proxies such as
42 national GDP and population statistics (Wada et al., 2011).

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The discretisation of functions into processes and subprocesses introduces a layer of complexity that leads astray from the elegance and simplicity of the global SOS framework. However, basin-scale water management is a complex task that encompasses technical challenges, trade-offs and conflicting interests (Mijic et al., 2024); therefore, we argue that a framework aimed at supporting a PB-inspired basin-management system must address this complexity to maintain the water system in a Holocene state at the basin scale. At its core, the proposed framework follows in the footsteps of Gleeson et al. 2020a,b and Zipper et al., 2020, but revises their approach by formulating the water boundary around water functions rather than stores to also acknowledge also the socio-economic dimension of the water system and stakeholders' participation, two element of water management that can't be discounted (Graversgaard et al., 2017, Hall et al., 2016) and require a further effort compared to the modelling-based PB global estimation. Nevertheless, stakeholder consultation and use of models are common practice also in other frameworks (Lim et al., 2022; Candido et al., 2022); therefore, from the operational point of view, the main novelty proposed by this framework is how the goals of the basin management plans are formulated rather than how the data are generated.

Despite the complexity of using water functions and creating a complex hierarchy, we claim that defining water boundaries around functions has a stronger biophysical and socio-ecological value than using stores, because "water volume" does not necessarily imply a functioning system. For example, considering only environmental flow requirements overlooks other important aspects, such as connectivity or water quality. The approach, therefore, has a strong link with ecosystem functionalities, as advocated in previous works (Ferretto et al. 2022), and assumes that to live within the doughnut, the water functions underpinning the ecosystem services on which humans depend must be maintained.

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Disclaimer

During the preparation of this work the author(s) used Grammarly Premium in order to spell-check and improve the readability of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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